Titanic Survival Data Analysis

Machine Design Document(MDD)

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Domain: Data Science & Machine Learning
Project Type: Exploratory Data Analysis (EDA)

Dataset: Titanic Passenger Dataset

Abstract

The Titanic Survival Analysis project aims to explore and analyze the Titanic passenger dataset to understand the factors that influenced survival during the Titanic disaster of 1912. The project applies data preprocessing, exploratory data analysis, visualization techniques, and machine learning concepts to uncover patterns in the data.

The analysis focuses on variables such as gender, passenger class, age, fare, and family size to determine how these features impacted survival rates. Data cleaning techniques were used to handle missing values, transform features, and prepare the dataset for analysis.

Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, the dataset was analyzed through visualizations including histograms, count plots, box plots, and heatmaps. These visual tools helped identify important relationships between different features and survival probability.

The project demonstrates how data science techniques can be applied to real-world datasets to derive meaningful insights.

Introduction

The sinking of the RMS Titanic remains one of the most tragic maritime disasters in history. The ship struck an iceberg during its maiden voyage in April 1912 and sank in the North Atlantic Ocean. Out of approximately 2,224 passengers and crew, more than 1,500 lost their lives.

Although the disaster occurred more than a century ago, the dataset collected from Titanic passengers provides a valuable opportunity for data analysis and machine learning. The dataset includes information such as passenger class, gender, age, ticket fare, cabin, and family members traveling together.

This project focuses on analyzing the dataset to identify patterns that influenced survival. The study applies modern data analysis tools to examine how social class, gender, age, and other attributes impacted survival chances.

Problem Statement

The Titanic dataset contains information about passengers who survived and those who did not. However, the dataset contains multiple variables that may influence survival. The problem is to analyze these variables and determine which features significantly affected survival probability.

The main challenge is understanding the relationships between variables such as passenger class, age, gender, and family structure and identifying patterns within the data.

The project aims to answer key analytical questions including:

- Did gender affect survival?
- Did passenger class influence survival rates?
- Were children more likely to survive?
- Did family size play a role in survival?

Objectives

The primary objectives of the project are:

- To understand the Titanic dataset and its structure.
- To perform data preprocessing and cleaning.
- To handle missing values in the dataset.
- To conduct exploratory data analysis (EDA).
- To visualize patterns and relationships in the data.
- To identify the most important factors affecting survival.
- To demonstrate practical applications of data science techniques.

Dataset Description

The Titanic dataset contains detailed information about passengers aboard the Titanic. Each row represents a passenger, and each column represents a feature describing that passenger.

Important features include PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked. Passenger class (Pclass) indicates socio-economic status, with first class being the highest. The Survived column is the target variable indicating whether the passenger survived the disaster.

Tools and Technologies

Python Programming Language

Pandas for data manipulation

NumPy for numerical computations

Matplotlib for visualization

Seaborn for statistical plotting

Jupyter Notebook for development and experimentation

These tools provide powerful capabilities for analyzing large datasets and creating visual insights.

Methodology

The methodology followed in this project includes several stages.

1. Data Collection

The Titanic dataset was obtained from publicly available datasets used in machine learning competitions.

2. Data Cleaning

Missing values were handled and irrelevant columns were inspected.

3. Feature Engineering

New features such as family size and age groups were created to improve analysis.

4. Exploratory Data Analysis

Different plots and charts were generated to understand relationships between variables.

5. Visualization

Graphs were used to represent patterns and trends in the dataset.

Data Preprocessing

Data preprocessing is an essential step in data analysis. The Titanic dataset contains missing values in multiple columns such as Age, Cabin, and Embarked.Missing Age values were replaced with the median age. Cabin values were replaced with 'Unknown'. Embarked values were filled using the most common port of embarkation.

Data types were checked and converted where necessary to ensure proper analysis.

Visualization

Visualization techniques were used to better understand the dataset.

Histograms were used to analyze age distribution.

Count plots were used to analyze gender distribution.

Box plots were used to detect outliers in age and fare.

Heatmaps were used to identify missing values.

These visualizations provided clear insights into survival trends.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of analyzing datasets to summarize their main characteristics using statistical and visualization techniques. It helps in understanding patterns, detecting anomalies, testing assumptions, and discovering relationships between variables.

In this project, several visualizations were used to analyze the Titanic dataset and understand the factors influencing passenger survival.

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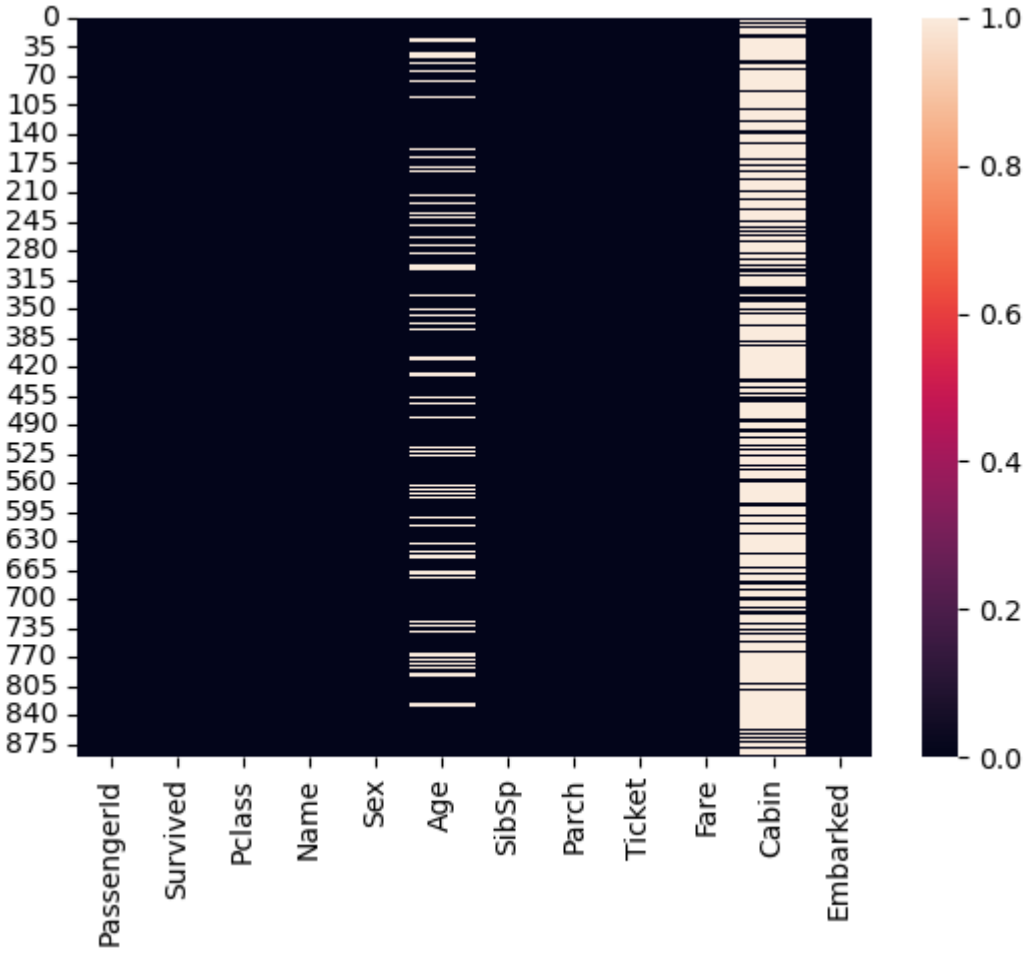
1. Missing Values Heatmap



Code Used

```
sns.heatmap(data.isnull())
```

Explanation

The heatmap visualization is used to identify missing values in the dataset. Each column of the dataset is displayed along the horizontal axis, and each row represents a passenger.

Bright or colored areas indicate missing values in specific columns.

Observation

From the heatmap, it was observed that:

- The **Age column contains several missing values**
- The **Cabin column contains a large number of missing entries**
- The **Embarked column has very few missing values**

Conclusion

Handling missing values is necessary before performing analysis or building predictive models.

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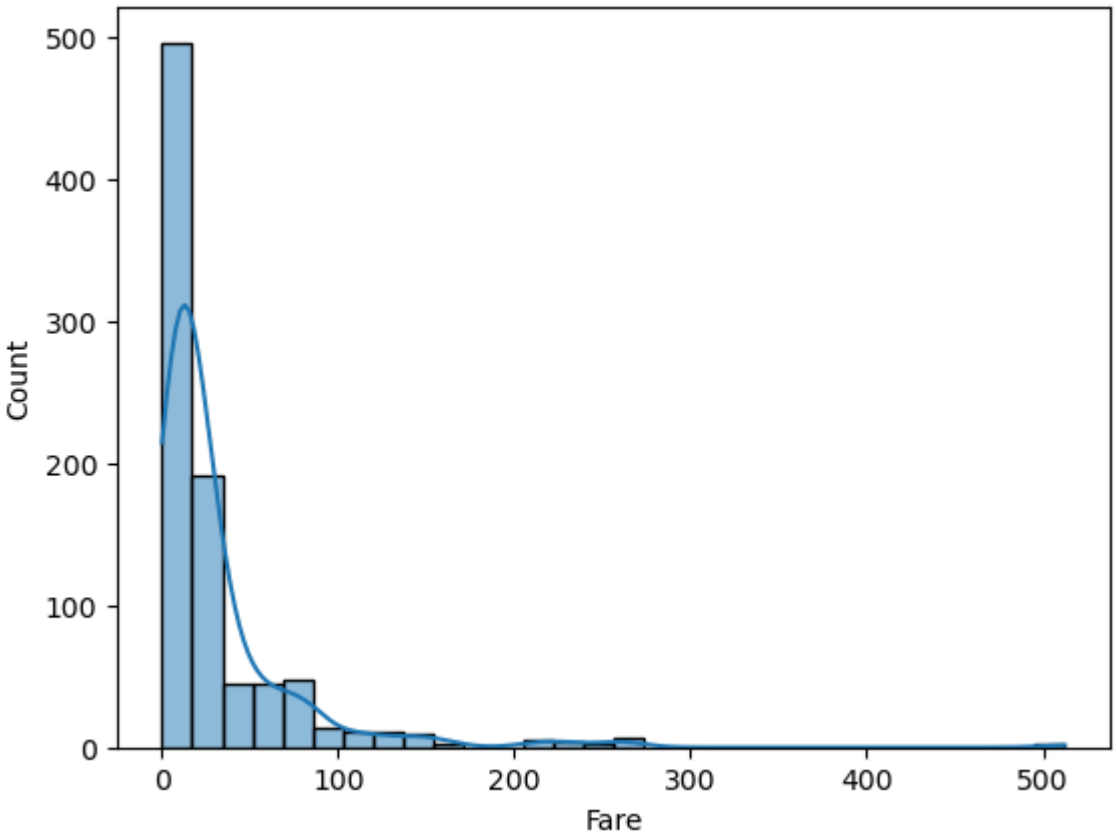
2. Fare Distribution



Code Used

```
sns.histplot(data['Fare'], bins=30,kde=True)
plt.show()
```

Explanation

This histogram shows the distribution of ticket prices paid by passengers.

Observation

The fare distribution is **highly skewed**:

- Many passengers paid **low fares**
- Few passengers paid **very high fares**

Conclusion

Ticket fare reflects **passenger class differences**

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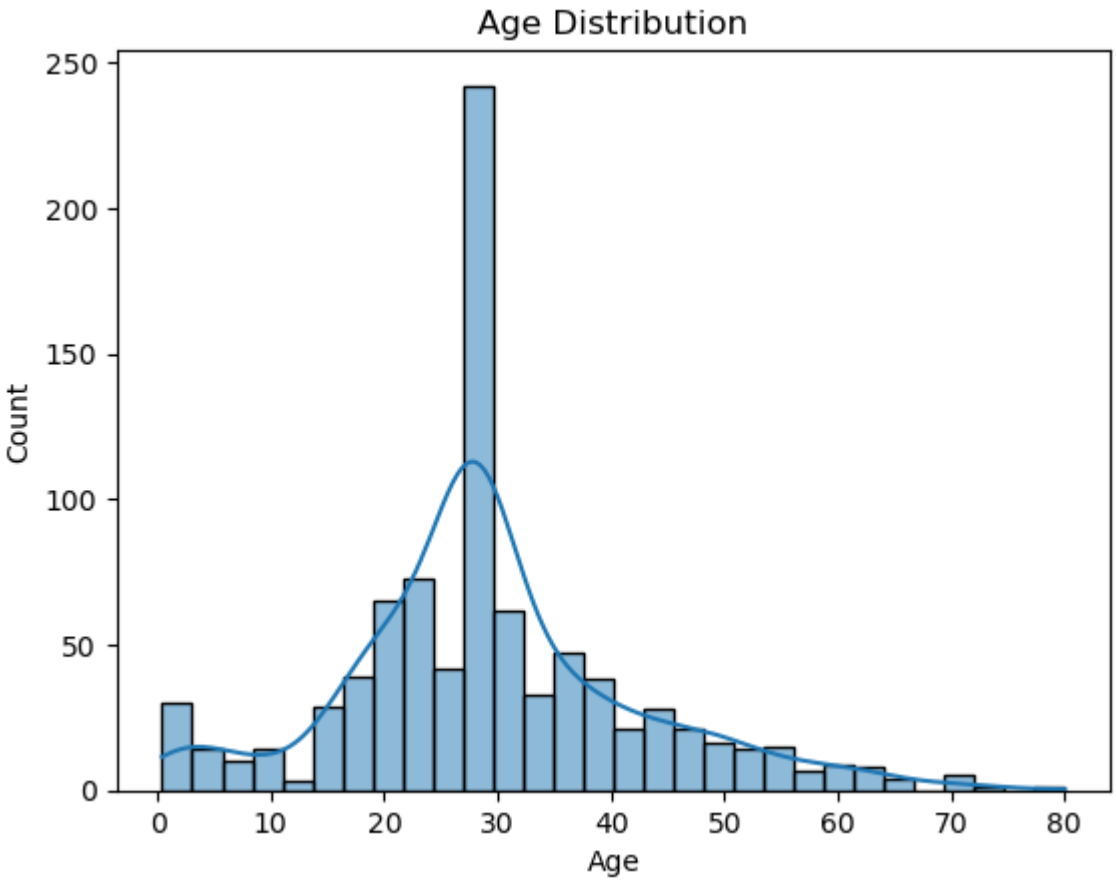
3. Age Distribution



Code Used

```
sns.histplot(data['Age'], bins=30,kde=True)

plt.title("Age Distribution")

plt.show()
```

Explanation

A histogram is used to analyze the distribution of passenger ages.

The KDE (Kernel Density Estimate) line helps visualize the probability density of age distribution.

Observation

The graph shows that:

- Most passengers were **between 20 and 40 years old**
- There were fewer elderly passengers.

Conclusion

The Titanic passenger population was dominated by young adults.

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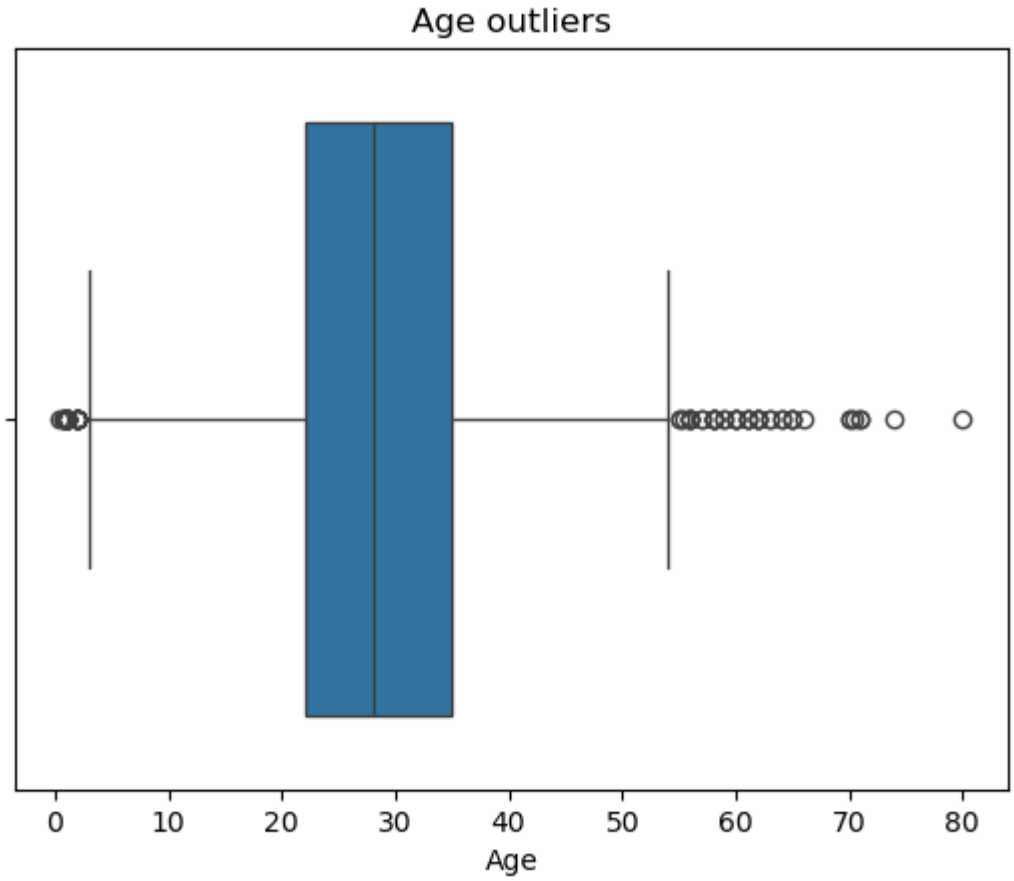
4. AGE OUTLIERS



Code Used

```
sns.boxplot(x=data['Age'])

plt.title('Age outliers')

plt.show()
```

Explanation

A boxplot is used to visualize the distribution of a numerical variable and identify outliers in the dataset.

In this graph:

- The X-axis represents the Age of passengers.

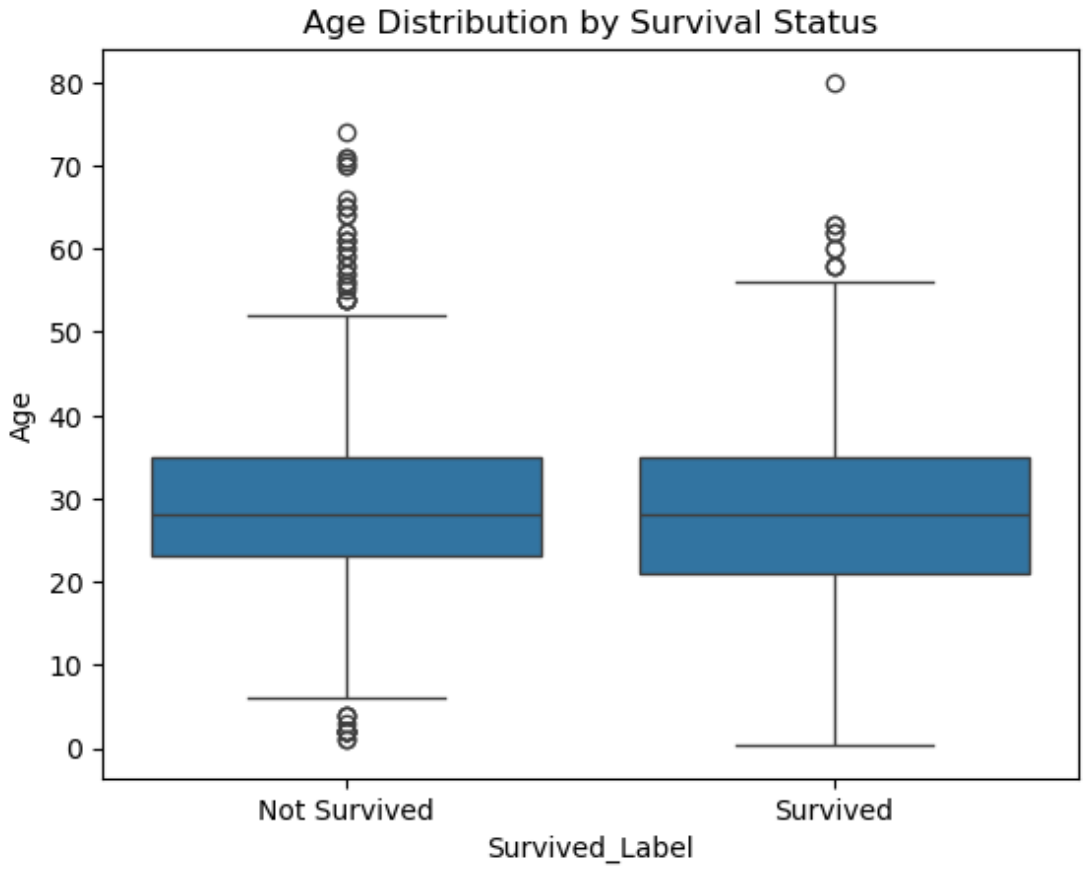
- The box shows the interquartile range (IQR), which contains the middle 50% of the data.
- The line inside the box represents the median age.
- The whiskers represent the range of typical values.
- Points outside the whiskers represent outliers.

This visualization helps in detecting unusual or extreme values in the age column.

Conclusion

The boxplot helps identify **age outliers in the dataset**. Detecting outliers is important because extreme values can affect statistical analysis and machine learning models. These values should be examined carefully during data preprocessing.

5.Age Distribution by Survival Status



Code Used

```
sns.boxplot(x='Survived_Label', y='Age', data=data)
plt.title('Age Distribution by Survival Status')
plt.show()
```

Explanation

A **boxplot** is used to visualize the distribution of a numerical variable across different categories.

In this graph:

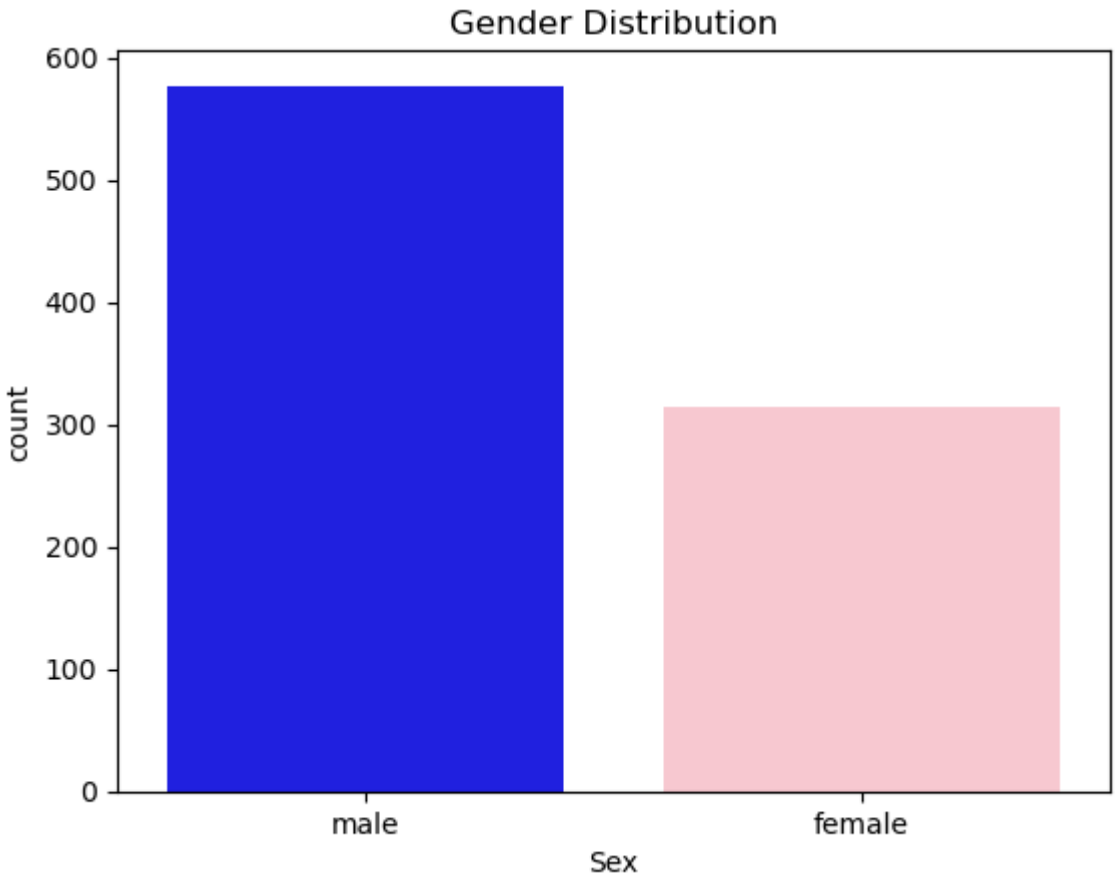
- **X-axis (Survived_Label)** represents survival status
    - Survived
    - Not Survived
- **Y-axis (Age)** represents the age of passengers.

The boxplot displays important statistical information such as:

- **Median (middle line in the box)**
- **Quartiles (Q1 and Q3)**
- **Minimum and maximum values**
- **Outliers (points outside the whiskers)**

This allows us to compare the **age distribution between passengers who survived and those who did not survive.**

6. Gender Distribution



Code Used

```
sns.countplot(
   x='Sex',
   hue='Sex',
   data=data,
   palette={'female': 'pink', 'male': 'blue'},
   legend=False
)

plt.title("Gender Distribution")
plt.show()
```

Explanation

This count plot shows the distribution of male and female passengers in the Titanic dataset.

Each bar represents the number of passengers belonging to each gender category.

Observation

The graph shows that:

- The number of **male passengers is significantly higher than female passengers**.

Conclusion

This indicates that more men were traveling on the Titanic than women.

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7. Survival Distribution

Code Used

```
data['Survived'].value_counts()
```

Output

```
0    549
1    342
```

Explanation

The value_counts() function counts the number of passengers who survived and those who did not survive.

Observation

- **549 passengers did not survive**
- **342 passengers survived**

Conclusion

More passengers died than survived during the Titanic disaster.

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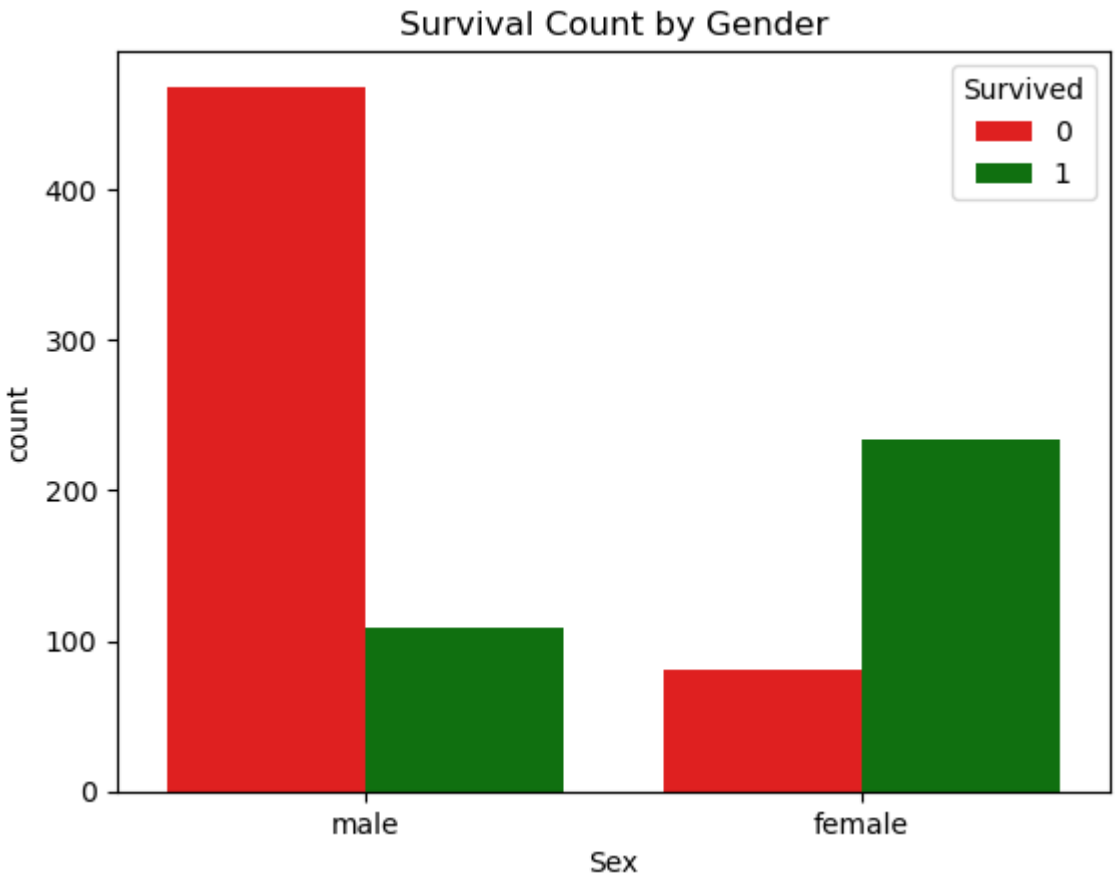
8. Survival by Gender



Code Used

```
sns.countplot(x='Sex',hue='Survived',data=data,palette={0: 'red', 1: 'green'})

plt.title('Survival Count by Gender')

plt.show()
```

Explanation

This graph compares survival counts between male and female passengers.

Two colors represent:

- Survived
- Not Survived

Observation

The visualization clearly shows that:

- **Most women survived**

- **Most men did not survive**

Conclusion

This confirms the historical rescue protocol **"Women and children first."**

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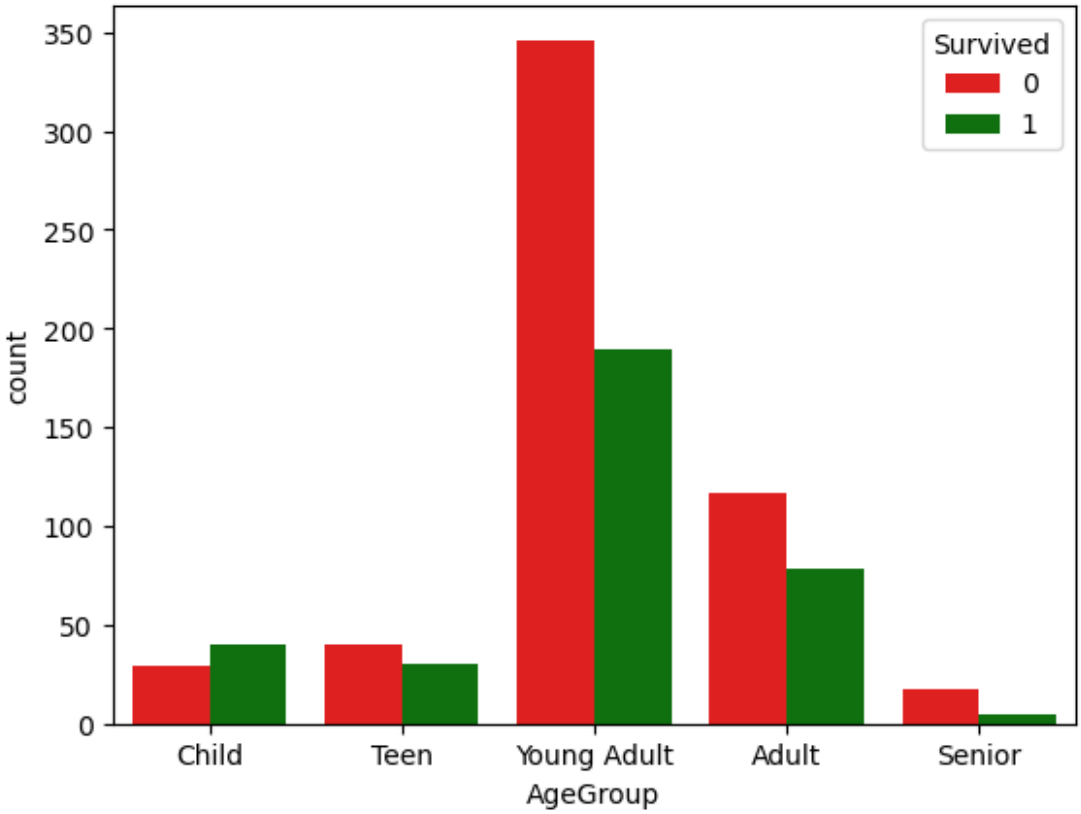
9. Survival by Age Group



Code Used

```
bins = [0, 12, 18, 35, 60, 100]
labels = ['Child', 'Teen', 'Young Adult', 'Adult', 'Senior']

data['AgeGroup'] = pd.cut(data['Age'], bins=bins, labels=labels)

sns.countplot(x='AgeGroup', hue='Survived', data=data, palette={0:'red',1:'green'})
plt.show()
```

Explanation

Passengers were grouped into age categories to analyze survival patterns across age groups.

Observation

- Children and young adults had relatively higher survival counts.
- Older passengers had fewer survival cases.

Conclusion

Age influenced survival probability, with younger passengers having better survival chances.

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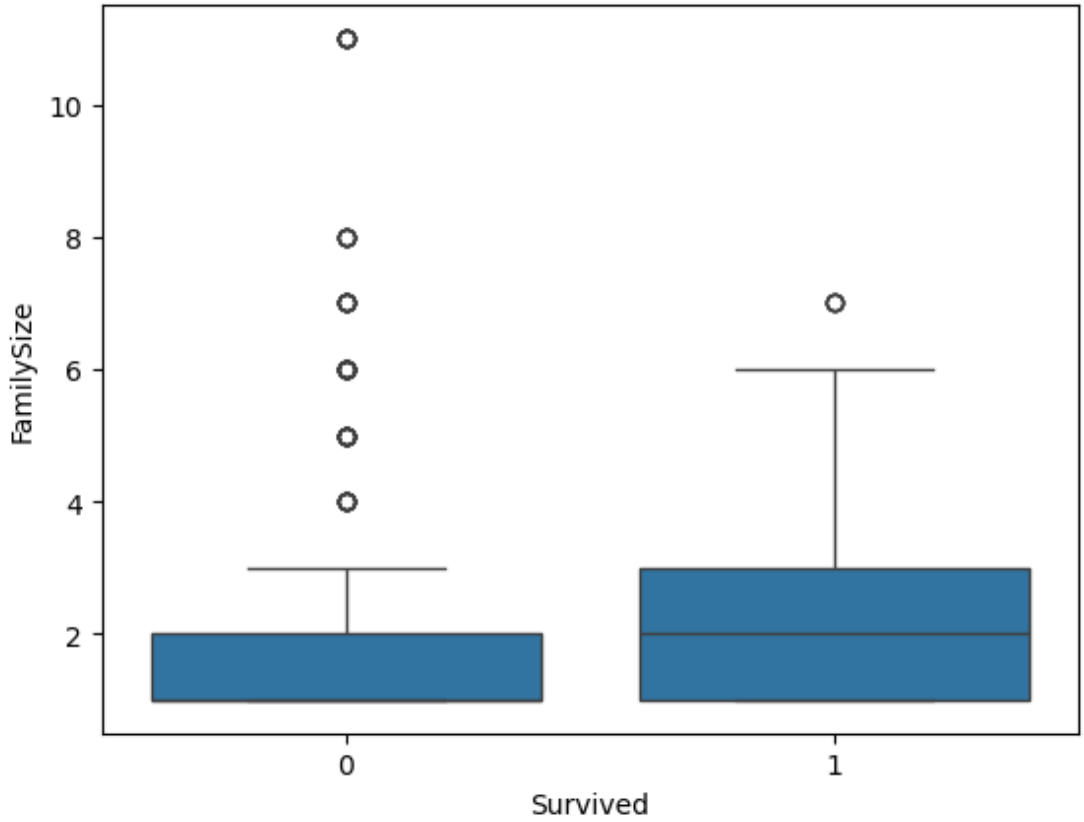
10. Survival by Family Size



Code Used

```
data['FamilySize'] = data['SibSp'] + data['Parch'] + 1

sns.boxplot(x='Survived', y='FamilySize', data=data)

plt.show()
```

Output

```
FamilySize
1    0.30
2    0.55
```

3 0.57
4 0.72
5 0.20
6 0.13
7 0.33
8 0.00
11 0.00

Explanation

Family size represents the total number of family members traveling together.

Observation

Passengers traveling with **small families (2–4 members)** had higher survival rates.

Conclusion

Passengers with moderate family sizes were more likely to survive compared to those traveling alone or in very large groups.

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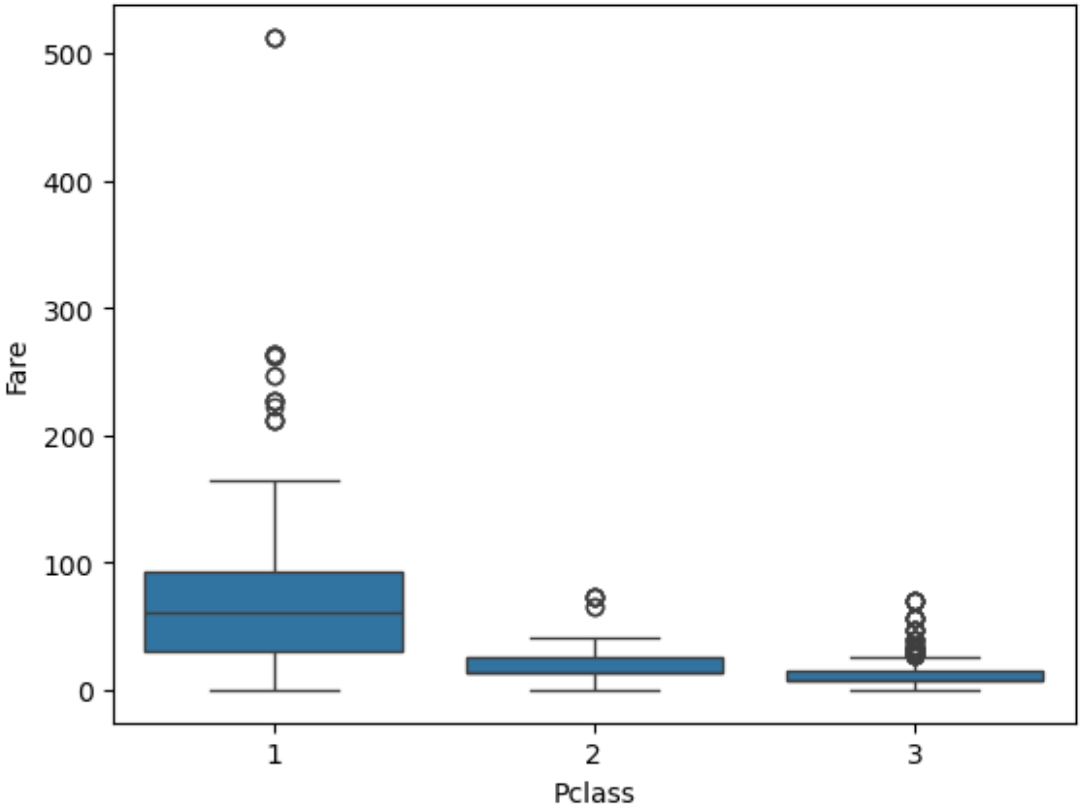
11. Fare Distribution by Passenger Class



Code Used

```
sns.boxplot(x='Pclass', y='Fare', data=data)
plt.show()
```

Explanation

This boxplot shows how ticket fares vary across different passenger classes.

Observation

- First-class passengers paid the highest fares.
- Third-class passengers paid the lowest fares.

Conclusion

Ticket fare reflects the socio-economic status of passengers.

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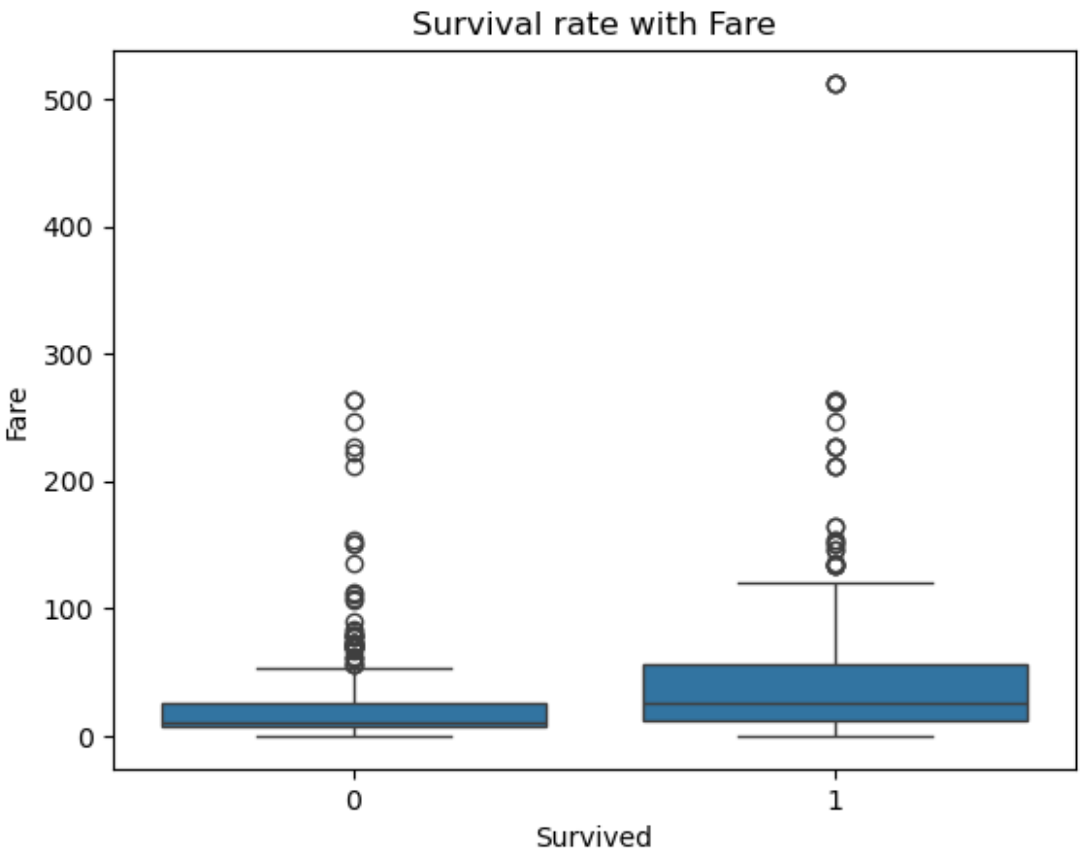
12. Survival by Fare



Code Used

```
sns.boxplot(y=data['Fare'], x=data['Survived'], data=data)
plt.title('Survival Rate with Fare')
plt.show()
```

Explanation

This boxplot compares the fare distribution between survivors and non-survivors.

Observation

Passengers who paid higher fares had a higher survival probability.

Conclusion

Passengers from higher economic classes had better survival chances.

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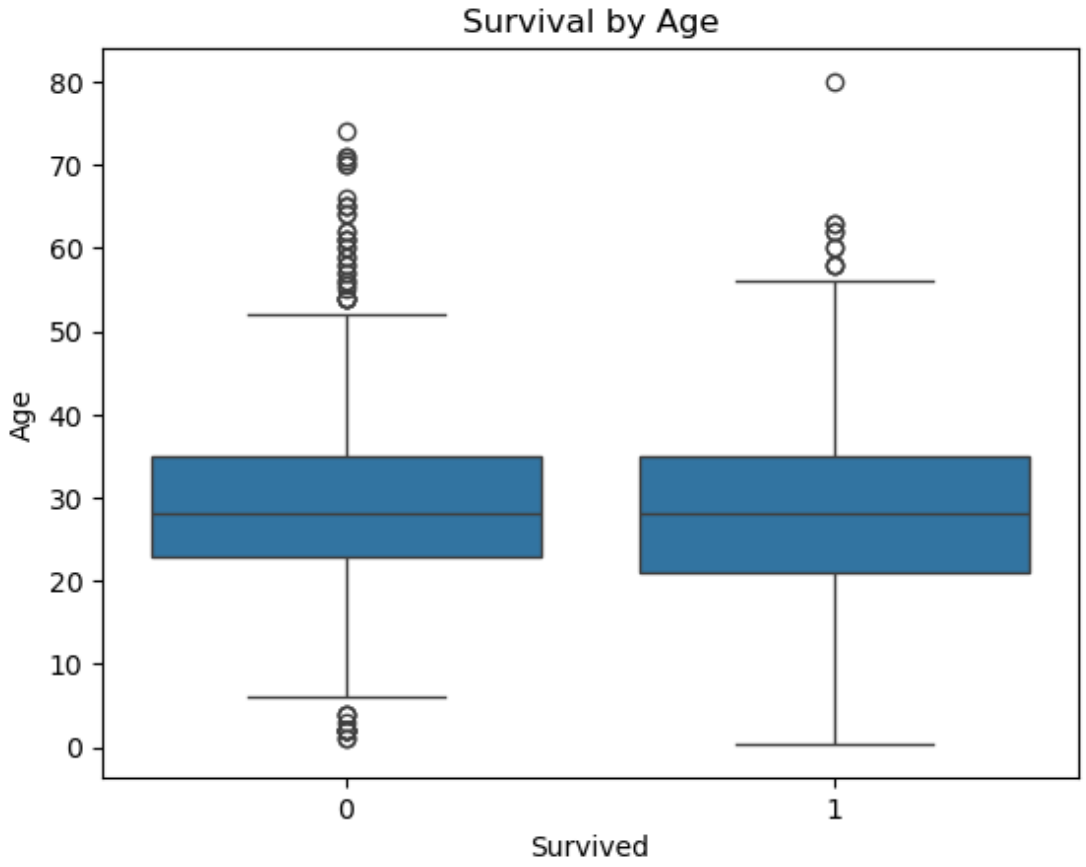
13. Survival by Age



Code Used

```
sns.boxplot(y=data["Age"], x=data['Survived'], data=data)
plt.title('Survival by Age')
plt.show()
```

Explanation

This boxplot compares age distribution between survivors and non-survivors.

Observation

Younger passengers appear more frequently among survivors.

Conclusion

Age played a role in survival probability.

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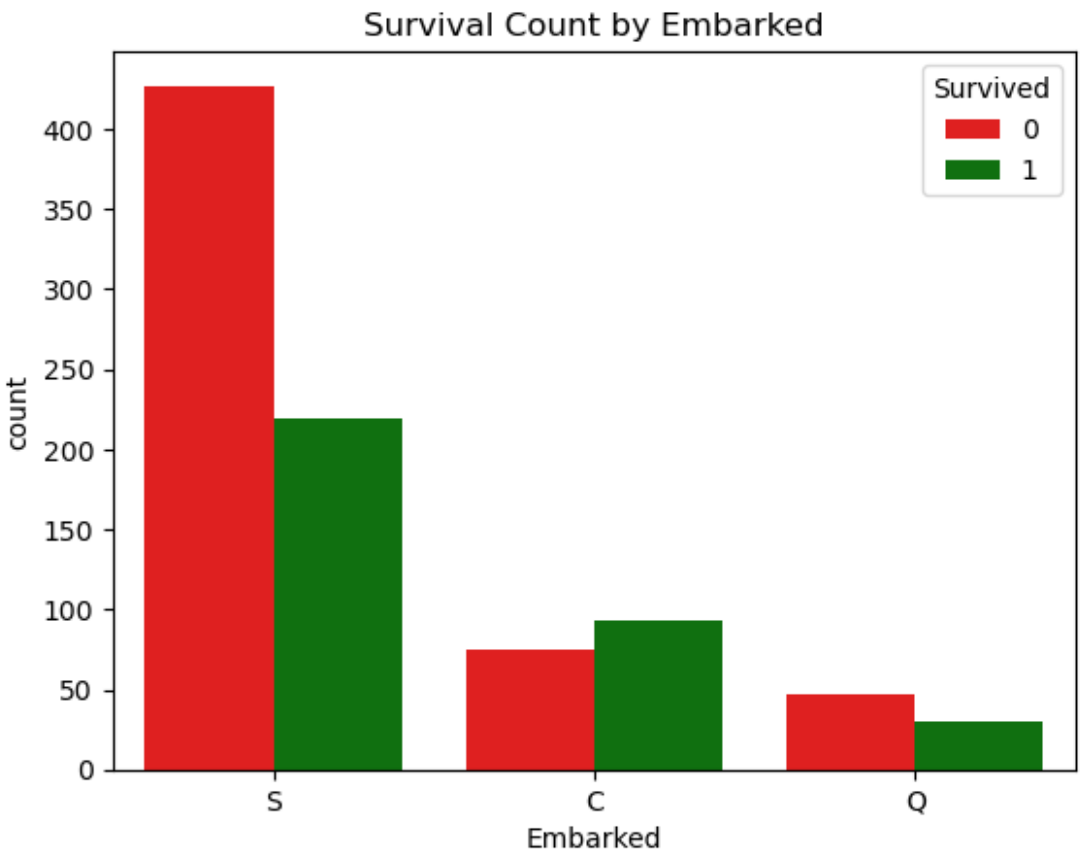
14. Survival by Embarkation Port



Code Used

```
sns.countplot(x=data["Embarked"], hue=data['Survived'], data=data,
palette={0:'red',1:'green'})
plt.title('Survival Count by Embarked')
```

Explanation

This visualization shows survival counts based on the port where passengers boarded the Titanic.

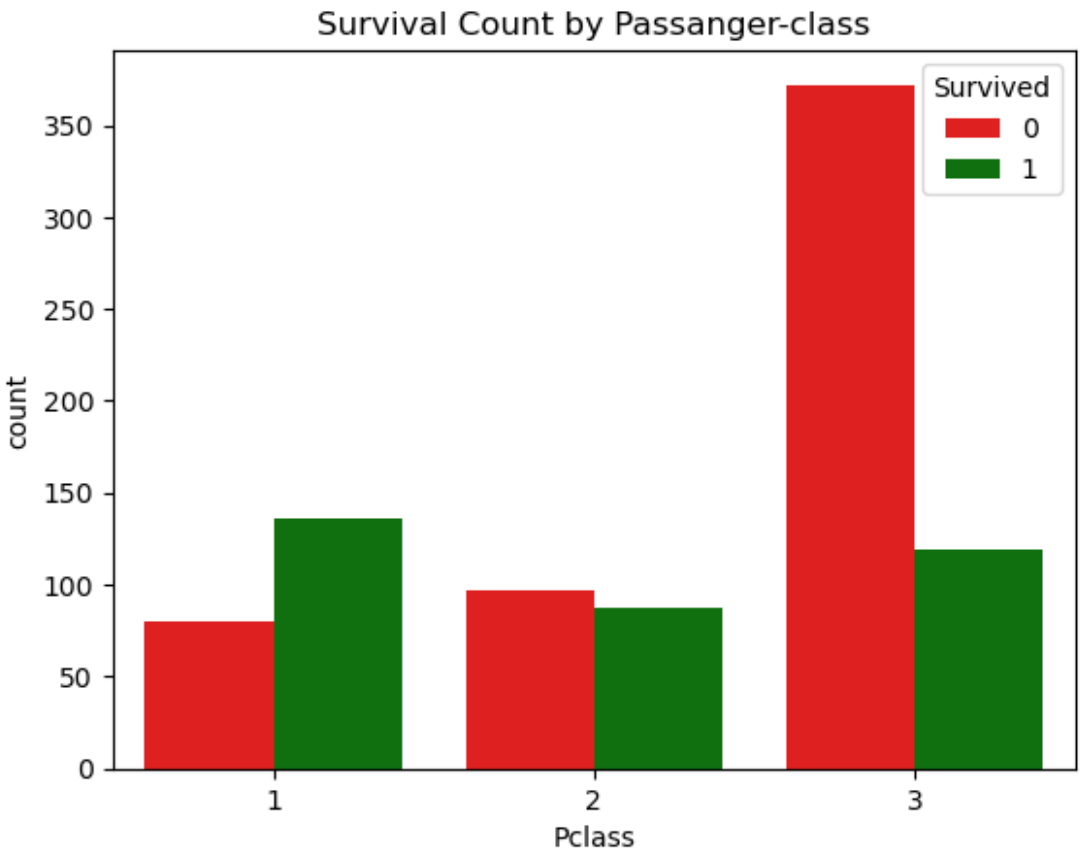
Observation

Different embarkation ports show varying survival rates.

Conclusion

Passenger origin location influenced survival patterns.

15. Survival by Passenger Class



Code Used

```
sns.countplot(x=data["Pclass"], hue=data['Survived'], data=data,
palette={0:'red',1:'green'})
plt.title('Survival Count by Passenger Class')
plt.show()
```

Explanation

This graph compares survival counts across passenger classes.

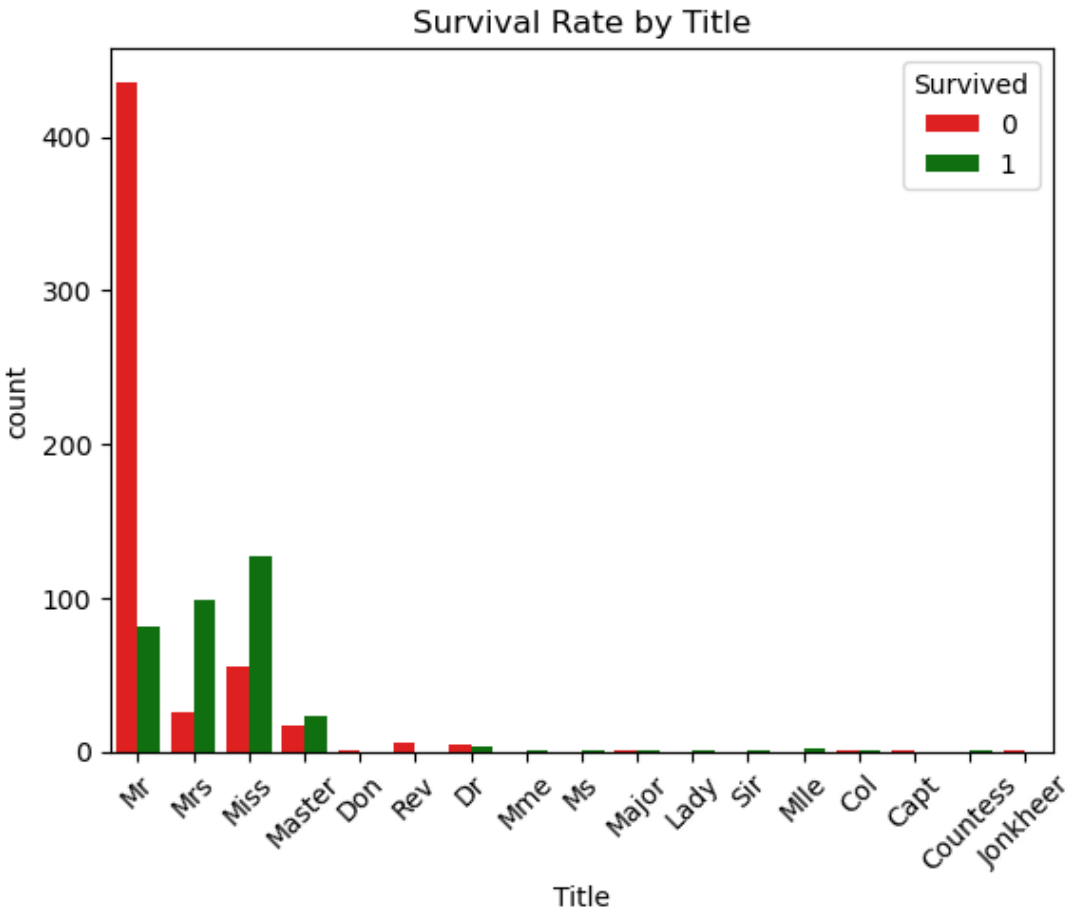
Observation

- First-class passengers had the highest survival rate.
- Third-class passengers had the lowest survival rate.

Conclusion

Passenger class strongly influenced survival probability.

16. Survival by Title



Code Used

```
data['Title'] = data['Name'].str.extract(' ([A-Za-z]+)\.', expand=False)

sns.countplot(x='Title', hue='Survived', data=data, palette={0:'red',1:'green'})
plt.xticks(rotation=45)
plt.title("Survival Rate by Title")
plt.show()
```

Explanation

Titles extracted from passenger names were analyzed to see survival differences among social titles.

Observation

Titles such as **Mrs and Miss show higher survival rates**.

Conclusion

Social roles and gender influenced survival probability.

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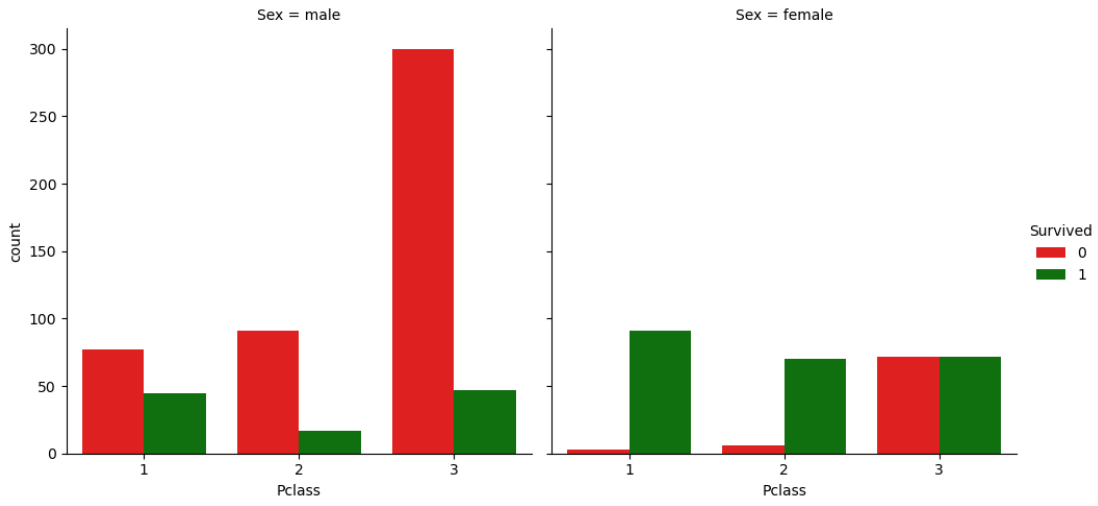
11. Passenger Class and Gender Survival



Code Used

```
sns.catplot(x='Pclass', hue='Survived', col='Sex', kind='count', data=data,
palette={0:'red',1:'green'})
plt.show()
```

Explanation

This visualization compares survival across passenger classes separately for males and females.

Observation

Female passengers had higher survival rates in every class.

Conclusion

Gender had a stronger influence on survival than passenger class.

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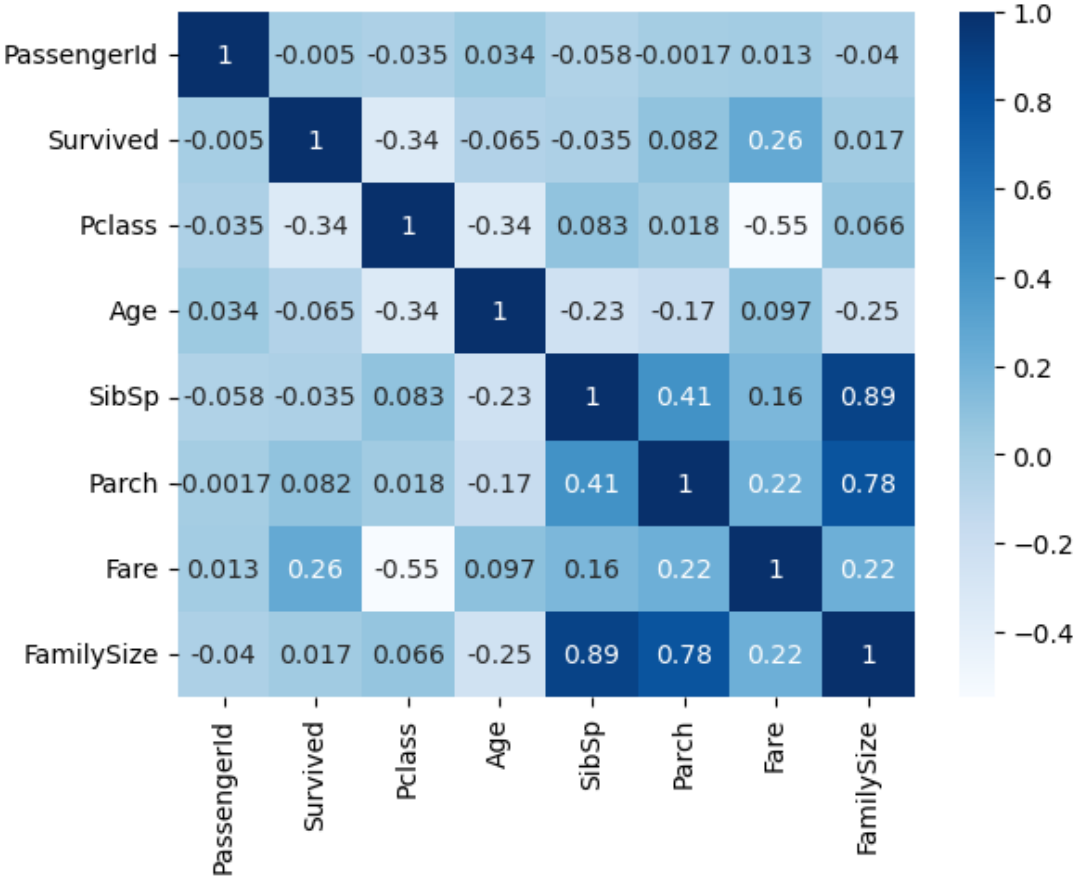
12. Correlation Heamap



Code Used

```
corr = data.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap='Blues')
plt.show()
```

Explanation

A correlation heatmap shows relationships between numerical features.

Observation

- Survival is positively correlated with **Fare**.
- Survival is negatively correlated with **Passenger Class**.

Conclusion

Passengers who paid higher fares and traveled in higher classes had better survival chances.

Final Insight from EDA

The EDA analysis revealed several important insights:

1. Women had significantly higher survival rates.
2. First class passengers had better survival chances.
3. Most passengers were young adults.
4. Fare and passenger class were strongly related.
5. Family structure influenced survival probability.

Results and Findings

The analysis revealed several important patterns.

Women had the highest survival rates.

Children had better chances of survival compared to adults.

Passengers in first class had significantly higher survival probability.

Passengers traveling with small families were more likely to survive.

These findings suggest that social class and rescue priorities played major roles during evacuation.

Future Scope

The project can be extended by applying machine learning models such as logistic regression, decision trees, and random forests to build predictive models.

A web dashboard can also be developed to visualize the analysis interactively. Additional datasets could further improve the model's predictive capabilities.

Conclusion

This project demonstrated how data science techniques can be used to analyze real-world datasets. By cleaning, processing,

and visualizing the Titanic dataset, valuable insights were obtained about the factors affecting survival.

The analysis shows that gender, passenger class, age, and family size significantly influenced survival chances. The project

highlights the importance of data analysis in understanding historical events and extracting meaningful insights from data.